

CVE-2026-82222: GiveWP object-injection RCE validation lab

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Content

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README.md

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CVE-2026-82222 — GiveWP Marker-Only RCE Validation Lab

Security-research material for reproducing and validating CVE-2026-82222 in an isolated Docker lab.

For Direct RCE Poc and url scan use [CVE-2026-8222-RCE.py](#)

Verdict

Status: proven in the supplied lab

GiveWP 4.16.5.1 permits an initially unauthenticated attacker to persist a PHP object graph, revive it through GiveWP session handling, and execute a fixed marker command as

the WordPress web-server user.

Tested positive result:

```
GiveWP:    4.16.5.1
WordPress: 6.6.2
PHP:       8.1.30
Result:    /tmp/CVE-2026-82222-RCE-GETBAG created by www-data
```

The end-to-end result was also reproduced against pristine, non-instrumented GiveWP 4.16.5.1 source. GiveWP 4.16.7.2 blocked the HTTP carrier in the tested configuration and independently blocked the terminal gadget during a direct control.

This demonstrates command execution inside the WordPress container. It does not demonstrate root access, container escape, lateral movement, or host compromise.

Safety boundary

Use this repository only on systems you own or are explicitly authorized to test.

The supplied PoC is intentionally constrained:

- It executes only `touch /tmp/CVE-2026-82222-RCE-GETBAG`.
- It offers no arbitrary-command option.
- It creates no shell, callback, persistence, or privilege escalation.
- It refuses non-loopback targets unless the operator supplies the explicit

`--allow-authorized-non-loopback` flag.

- Docker publishes WordPress only on `127.0.0.1`.
- The Compose project name is derived from the checkout path so one clone cannot

tear down another clone's containers or volumes.

- The lab uses pristine official plugin source; it does not instrument or patch

the vulnerable target.

The HTTP test creates a disposable donor user, metadata, and GiveWP session rows. Use the included reset command after testing.

Quick start

Requirements:

- Docker with Compose v2
- Python 3.10 or later
- `curl`
- `unzip`
- `sha256sum` or `shasum`

- Network access to `downloads.wordpress.org` for official plugin archives

Run the complete vulnerable/patched matrix:

```
./lab verify
```

The command:

1. Downloads GiveWP 4.16.5.1 and 4.16.7.2 from WordPress.org.
2. Verifies both SHA-256 hashes.
3. Builds a fresh loopback-only WordPress 6.6.2/PHP 8.1 lab with normal WordPress registration explicitly disabled.

1. Tests pristine 4.16.5.1 and requires marker creation by the web user.
2. Builds a second fresh lab with pristine 4.16.7.2.
3. Requires the HTTP marker to remain absent.
4. Bypasses ingress in a direct control and confirms the patched

`ProviderForwarder` terminal also rejects the string callable.

The patched lab remains running at the end. Remove it with:

```
./lab reset
```

To use another loopback port:

```
LAB_PORT=8099 ./lab verify
```

Expected evidence

The vulnerable control must end with concrete terminal evidence:

```
[PASS] E1: unauthenticated registration issued auth cookie
[PASS] E3: serialized graph persisted in own last_name
[PASS] E4: donation-form nonce obtained
[PASS] E5: session write reached expected post-sink HTTP status=500
[PASS] E6: session read/destruction trigger completed
marker present and owned by the WordPress web user
RESULT: VULNERABLE CONTROL CONFIRMED
```

The patched control must show:

```
[PASS] P1: patched registration gate blocked auth cookie
DIRECT_MARKER=absent
HTTP marker absent and direct terminal gadget blocked
RESULT: PATCHED CONTROL CONFIRMED
```

An HTTP 500, stored payload, exception, or detector hit without the marker is not accepted as RCE proof.

Manual lab lifecycle

Start and test the vulnerable release:

```
./lab start vulnerable
./lab test
```

Start and test the patched release:

```
./lab start patched
./lab test
```

Inspect current state:

```
./lab status
```

Remove containers, volumes, test users, sessions, and marker state:

```
./lab reset
```

Cached plugin ZIPs and extracted assets are preserved for faster reruns. Remove those exact generated assets as well with:

```
./lab reset --purge-assets
```

Affected and tested versions

Version	Assessment
GiveWP 4.16.5.1	RCE reproduced end to end
GiveWP 4.16.6–4.16.7.1	Reported affected; not individually reproduced here
GiveWP 4.16.7.2	Patched negative controls reproduced

Version	Assessment
Later releases	Not individually tested; update to the latest supported release

The public advisory identifies releases through 4.16.7.1 as affected. This repository directly proves only the two versions in its positive/negative test matrix.

References:

- [Patchstack advisory](#)
- [CVE-2026-82222](#)
- [GiveWP hardening commit](#)
- [Official GiveWP plugin page](#)

Root cause

The exploit combines several behaviors:

1. GiveWP 4.16.5.1 exposes a registration action that creates and authenticates a low-privilege donor even when normal WordPress registration is disabled.

1. That user can persist serialized data in their own name metadata.

2. `Give\Helpers\Utils::maybeSafeUnserialize()` uses

`allowed_classes => false`, producing `__PHP_Incomplete_Class`, but a later serialization preserves the original class names and properties.

1. The graph is stored in a GiveWP purchase session.

2. A later unrestricted `maybe_unserialize()` revives the shipped classes.

3. Automatic destruction enters a complete POP chain to `system()`.

Four literal namespace backslashes are required in the HTTP carrier. The two effective stripping passes reduce them `4 -> 2 -> 1`. The payload is NUL-free, and PHP 8.1.30 was verified to hydrate the plain serialized name for the private `Session::$attributeName` property.

Complete POP chain

```

TCPDF::__destruct()
-> TCPDF::_destroy(true)
-> foreach ($this->imagekeys as $file)
-> Symfony Session::getIterator()
-> Session::getAttributeBag()
-> Session::getBag($this->attributeName)
-> $this->storage->getBag($attributeName)
-> DonationFactory->__call('getBag', [$attributeName])
-> call_user_func_array('system', [$attributeName])
-> system('touch /tmp/CVE-2026-82222-RCE-GETBAG')

```

Attacker-controlled graph:

```

TCPDF
├─ file_id = unique request identifier
├─ imagekeys = Give\Vendors\Symfony\Component\HttpFoundation\Session\Session
│   └─ attributeName = fixed marker command
│       └─ storage = Give\TestData\Factories\DonationFactory
│           └─ loadedProviders["getBag"] = "system"

```

The critical hidden transition is PHP's implicit `IteratorAggregate` dispatch.

`TCPDF::$imagekeys` is untyped, so assigning a `Symfony Session` causes `foreach` to invoke `Session::getIterator()`.

The marker command executes before Symfony enforces the `getAttributeBag(): AttributeBagInterface` return type. The resulting `TypeError` and HTTP 500 are post-sink effects.

What earlier attempts missed

The rejected candidate graph used:

```

TCPDF::$objcopy
-> DonationFactory::$loadedProviders['__destruct'] = 'system'

```

That graph cannot work. `TCPDF::_destroy()` only unsets `objcopy`; PHP does not route automatic destruction through `__call('__destruct', ...)`.

The missing operation was:

```

foreach ($this->imagekeys as $file) {

```

The earlier review followed destructors and explicit method calls but did not recursively inspect the implicit object protocol triggered by `foreach`. `Symfony Session` supplies the

missing bridge:

- `foreach` invokes `getIterator()`.
- `getIterator()` reaches `getBag($attributeName)`.
- Attacker-controlled `storage` is a `DonationFactory`.
- The undefined `getBag` invokes `ProviderForwarder::__call()`.
- `loadedProviders['getBag'] = 'system'` selects the callable.
- `attributeName` supplies the command argument.

Source evidence

Relevant GiveWP 4.16.5.1 locations:

Component	Location
Guarded initial unserialize	<code>src/Helpers/Utils.php:203-217,237-241</code>
Donation carrier	<code>includes/process-donation.php:157</code>
GiveWP session persistence	<code>includes/class-give-session.php:364-368,489-508</code>
Unrestricted session revival	<code>includes/class-give-session.php:347-350</code>
TCPDF destructor	<code>vendor/tecnickcom/tcpdf/tcpdf.php:2050-2052</code>
Attacker-controlled iteration	<code>vendor/tecnickcom/tcpdf/tcpdf.php:7885-7907</code>
Symfony iterator bridge	<code>vendor/vendor-prefixed/symfony/http-foundation/Session/Session.php:134-136</code>
Symfony <code>getBag</code> bridge	<code>vendor/vendor-prefixed/symfony/http-foundation/Session/Session.php:259-283</code>
Terminal callable	<code>src/TestData/Framework/ProviderForwarder.php:19-23</code>
GiveWP Composer autoloaders	<code>give.php:624-625</code>

Official release hashes:

```
give.4.16.5.1.zip
95fc6709b6ed284074bf09be096e28d6299fd4a103848c2b1c3bf8a5772cf98c

give.4.16.7.2.zip
c5bc98da2abb748c64d31a43679ff5f223092dfff6d214132a55dcbc948e91ae
```

The lab re-extracts each plugin from its verified ZIP before every fresh start, copies it into WordPress without modification, and byte-compares the deployed complete plugin file tree with the verified source. The three official Docker base images are also pinned to their multi-architecture manifest digests.

Why 4.16.7.2 blocks the chain

GiveWP 4.16.7.2 adds independent defenses across registration, serialized input handling, session revival, data migration, and gadget hardening.

The terminal hardening requires the resolved object to implement the expected provider contract:

```
if ( ! $provider instanceof Contract\Provider ) {  
    return null;  
}
```

The attacker-controlled string `system` is therefore rejected. The direct control in this repository bypasses the HTTP carrier and verifies that this terminal defense alone leaves its marker absent.

Checking a real deployment

A negative PoC result by itself does **not** prove a site is patched. A WAF, different routing, disabled registration, missing legacy form, session configuration, or disabled PHP command functions can all prevent the marker on a still-vulnerable codebase.

Preferred validation procedure:

1. Record the deployed GiveWP version and source revision.
2. Back up or snapshot the WordPress site and database.
3. Restore that snapshot into an isolated staging environment.
4. Disable unnecessary outbound network access.
5. Run this marker-only verifier against the clone.
6. Update GiveWP to the latest supported release, with 4.16.7.2 as the minimum release containing the tested fixes.

1. Repeat the identical test and require marker absence.
2. Independently verify the installed plugin version and patched source.

Version check:

```
wp plugin get give --fields=name,status,version
```

Remediation and incident review

Update GiveWP to the latest supported version. After updating:

- Clear PHP opcode caches where applicable.
- Confirm no older GiveWP copy remains active or web-accessible.
- Review unexpected WordPress or donor accounts.
- Search user metadata and GiveWP sessions for serialized `TCPDF`, `Symfony`

`Session`, or `loadedProviders` graphs.

- Correlate registration, profile changes, donation requests, and subsequent HTTP 500 responses from the same session.

- Investigate unexpected web-server child processes or filesystem changes.

If an internet-exposed vulnerable installation contains object-injection artifacts, treat it as a potential compromise rather than only updating the plugin.

Repository layout

```
.
├── .github/workflows/validate.yml
├── .gitignore
├── README.md
├── SECURITY.md
├── docker-compose.yml
├── lab
├── poc
│   ├── direct-pop-control.php
│   └── poc.py
└── scripts
    └── fetch-assets.sh
```

Not committed:

```
official plugin ZIPs
extracted GiveWP source
instrumented target source
cookies or session data
runtime evidence/logs
internal target addresses or reusable credentials
obsolete objcopy payloads
historical validation output
```

Validation boundaries

Claim	Status
Persistent PHP object-injection carrier	Proven
Revival of shipped classes	Proven
Complete stock POP chain	Proven
Fixed marker executed as the web user	Proven
Reproduction against pristine 4.16.5.1	Proven
Negative controls against pristine 4.16.7.2	Proven
Every intermediate affected release tested	Not tested
Root privilege	Not claimed
Container escape	Not tested
Host compromise	Not tested

The proof standard is deliberately strict: only an observable command marker through the unmodified stock chain is labeled RCE. HTTP acceptance, serialization, exceptions, and detector hits are intermediate evidence.

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